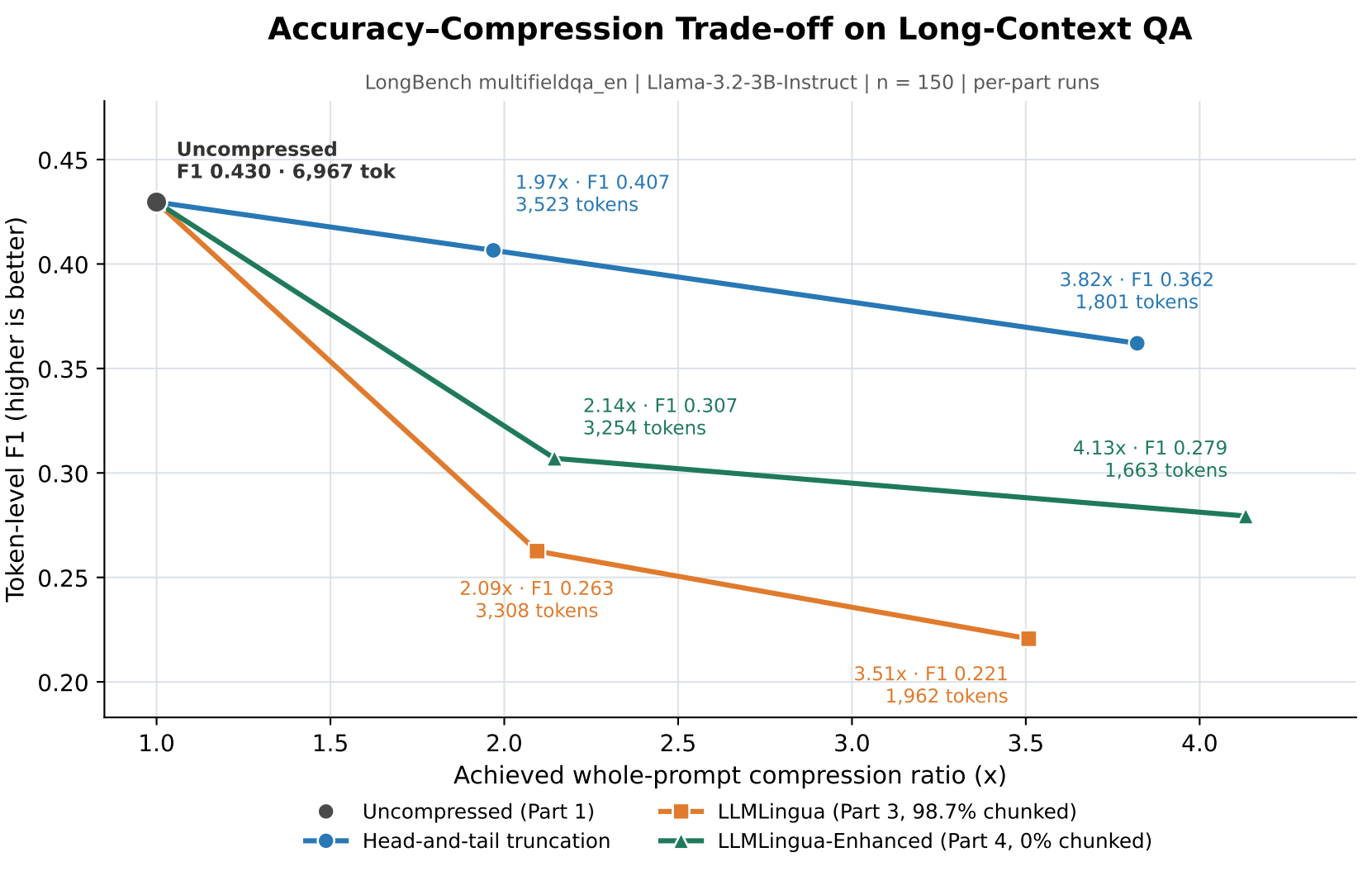
Accuracy–Compression Trade-off on Long-Context QA
LongBench multifieldqa_en | Llama-3.2-3B-Instruct | n = 150 | per-part runs
Uncompressed
F1 0.430 · 6,967 tok
1.97x · F1 0.407
3,523 tokens
3.82x · F1 0.362
1,801 tokens
2.14x · F1 0.307
3,254 tokens
4.13x · F1 0.279
1,663 tokens
2.09x · F1 0.263
3,308 tokens
3.51x · F1 0.221
1,962 tokens
Token-level F1 (higher is better)
0.45
0.40
0.35
0.30
0.25
0.20
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Achieved whole-prompt compression ratio (x)
Uncompressed (Part 1)
Head-and-tail truncation
LLMLingua (Part 3, 98.7% chunked)
LLMLingua-Enhanced (Part 4, 0% chunked)